19/05/2026

Code P

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 **Project Phoenix for Re-NEET-2026_PST-02P** **Time : 180 Min.**

Topics Covered:
Physics: Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves.
Chemistry: Organic Chemistry - Some Basic Principles and Techniques, Hydrocarbons, Thermodynamics, Classification of Elements and Periodicity in Properties, Redox Reactions.
Botany: Plant growth and development, Cell cycle and cell division, The living world, Morphology of flowering plants, Anatomy of flowering plants.
Zoology: Body Fluids and Circulation, Excretory Products and their Elimination, Locomotion and Movement, Neural Control and Coordination, Chemical Coordination and Integration.

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A wire of length *l* is stretched by a force *F* to cause an extension *x*. Another wire of same material and same volume but 50% larger in length than the first wire, the force required for the same extension will be

(1) $\frac{3}{2}F$

(2) $\frac{2}{3}F$

(3) $\frac{9}{4}F$

(4) $\frac{4}{9}F$

2. A cylindrical rod of radius ‘r’ on the ground supports a load ‘W’ and experiences a compression of ‘e’. If a cylindrical hole of radius r/2 is drilled coaxially and it now supports the same load, the compression will

(1) decrease by e/2

(2) increase by e/3

(3) increase by e/4

(4) decrease by 4e/3

3. The lower face of a steel slab of length 500 mm, breadth 400 mm and height 40 mm is kept fixed and a force of $10^5 N$ is applied on the upper face parallel to it. If the displacement of the upper face relative to the lower face is $4\times 10^{-3}\ mm$, the rigidity modulus of steel is

(1) $5\times 10^{6}\ N\,m^{-2}$

(2) $5\times 10^{8}\ N\,m^{-2}$

(3) $5\times 10^{10}\ N\,m^{-2}$

(4) $5\times 10^{9}\ N\,m^{-2}$

4. The value of poisson’s ratio when longitudinal and transverse strains are equal is

(1) –1

(2) 2

(3) –0.5

(4) 0

5. When a pressure of 200 atmosphere is applied on a spherical ball, then its volume is reduced by 0.02%. The bulk modulus of the material of the ball is

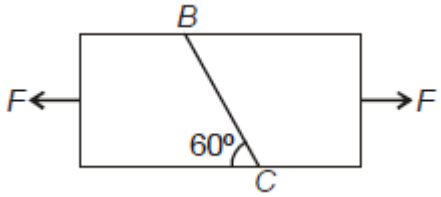
(1) 10^{11} N/m^2

(2) 10^{6} N/m^2

(3) 10^{4} N/m^2

(4) 10^{12} N/m^2

6. Two equal and opposite force of magnitude *F* act on a rod of uniform cross sectional area A as shown in the figure. Find the ratio of shearing stress and longitudinal stress on section *BC*.



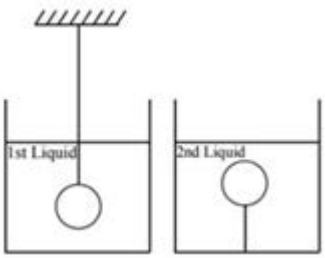
(1) $2:\sqrt{3}$

(2) $\sqrt{3}:2$

(3) $\frac{1}{\sqrt{3}}$

(4) $\sqrt{3}:1$

7. The figure shows two identical bodies submerged in two different liquids. What is the density of the spherical body if the tension in the strings have the same value?
$\left(\rho_1 = 800\,\text{kg}/\text{m}^3 \text{ and } \rho_2 = 1000\,\text{kg}/\text{m}^3\right)$



(1) 100 kg/m^3

(2) 200 kg/m^3

(3) 900 kg/m^3

(4) 600 kg/m^3

8. On which of the following concepts, deviation of a Spinning cricket ball is based?

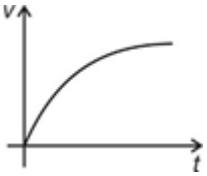
(1) Bernoulli's theorem

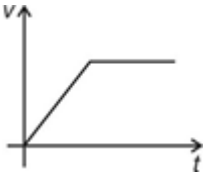
(2) Equation of continuity

(3) Surface tension

(4) Viscosity

9. Choose the correct graph (velocity-time) for a stone dropped in a viscous liquid having density less than stone.

(1) 

(2) 

(3) 

(4) 

10. A tank is filled to a height *H*. The range of water coming out of a hole which is at depth $\frac{H}{4}$ from the surface of water level is

(1) $\frac{H}{2}$

(2) $\frac{\sqrt{3}H}{2}$

(3) $\frac{\sqrt{3}H}{4}$

(4) $\sqrt{3}H$

11. Read the following statements and choose the correct option.
Statement A: Surface tension is a force per unit length acting in the plane of interface between the plane of liquid and any other substance.
Statement B: Surface tension is the extra energy that the molecules at interface have as compared to molecules in interior.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

12. The specific volume of water is minimum at

(1) 4 K

(2) 4°F

(3) 39.2 K

(4) 39.2°F

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13. A capillary tube of radius r is immersed in water and water rises in it to a height *h*. The mass of the water in the capillary is 5 g. Another capillary tube of radius 2*r* is immersed in water. The mass of water that will rise in this tube is:

(1) 5.0 g

(2) 10.0 g

(3) 20.0 g

(4) 2.5 g

14. Consider a water droplet having radius 10^{-2} m. It is broken into identical 1000 droplets. If surface tension of the water is 0.075 N/m then change in surface energy is

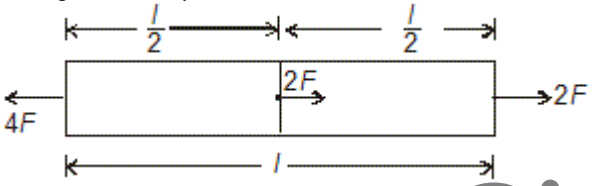
(1) $27\pi \times 10^{-5}$ J

(2) $30\pi \times 10^{-5}$ J

(3) $27\pi \times 10^{-3}$ J

(4) $30\pi \times 10^{-3}$ J

15. For the figure given below, the total elongation of rod is (Given *l* = length of rod, *A* = Area of cross section, *Y* = Young's modulus)



(1) $\frac{3Fl}{AY}$

(2) $\frac{8FI}{AY}$

(3) $\frac{FI}{AY}$

(4) $\frac{2FI}{AY}$

16. Heat required to convert 20 gram of ice at –20°C into steam at 100°C is

(1) 2400 cal

(2) 24000 cal

(3) 8640 cal

(4) 14600 cal

17. A black body calorimeter filled with hot water cools from 60°C to 50°C in 4 minutes and 40°C to 30°C in 8 minutes. The approximate temperature of surrounding is

(1) 10°C

(2) 15°C

(3) 20°C

(4) 25°C

18. The wavelength of radiations of maximum intensity radiated by a black body is proportional to its absolute temperature *T* as

(1) T

(2) T^{-1}

(3) T^{2}

(4) T^{-2}

19. The volume thermal expansion coefficient of an ideal gas at constant pressure is

(1) $\frac{1}{V}$

(2) $\frac{1}{T}$

(3) $\frac{P}{T}$

(4) $\frac{1}{\rho}$

20. A closed gas cylinder is divided into two parts by a piston held tight. The pressure and volume of gas in two parts respectively are (P, $5V$) and ($10P$, V). If now piston is left free and system undergoes isothermal process then volume of gas in two parts respectively are

(1) ($2V$, $4V$)

(2) ($3V$, $3V$)

(3) ($5V$, V)

(4) ($4V$, $2V$)

21. **Assertion (A) :** Real gases approach ideal gas behaviour at low pressure and high temperature.
Reason (R) : According to Boyle's law, pressure of a gas is inversely proportional to volume of a gas at constant temperature.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

22. The efficiency of Carnot engine working between 327°C and 27°C is

(1) 40%

(2) 50%

(3) 64%

(4) 34%

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23. During an adiabatic process, the pressure of a gas is found to be proportional to cube of its absolute temperature. Then $\frac{C_p}{C_v}$ for the gas is

(1) 2

(2) $\frac{4}{3}$

(3) $\frac{3}{2}$

(4) $\frac{5}{3}$

24. The quantities of heat required to raise temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5r_2$) through 1 K is in the ratio of

(1) $\frac{9}{8}$

(2) $\frac{27}{2}$

(3) $\frac{27}{8}$

(4) $\frac{19}{11}$

25. During an adiabatic expansion of 5 moles of gas, the internal energy decrease by 75 J. The work done during the process is

(1) Zero

(2) 15 J

(3) –75 J

(4) + 75 J

26. Match the following for work done in various thermodynamic processes. (Symbols have their usual meaning)

List-I (Process)	List-II (W.D.)
A. Isobaric expansion	(I) Zero
B. Isochoric	(II) $-\Delta U$
C. Adiabatic	(III) ΔQ
D. Isothermal	(IV) $nR\Delta T$

(1) A-IV, B-I, C-II, D-III

(2) A-IV, B-I, C-III, D-II

(3) A-IV, B-II, C-III, D-I

(4) A-I, B-II, C-III, D-IV

27. A mixture of 3 moles of helium gas and 2 moles of neon gas (atomic mass = 22 u) is kept at 127°C in a container. The ratio of their rms speeds is

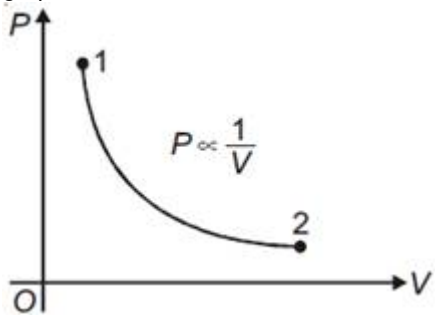
(1) 24 m/s

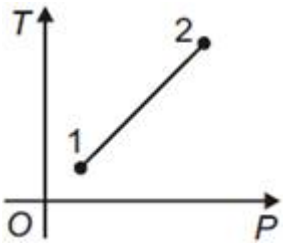
(2) 4.3 m/s

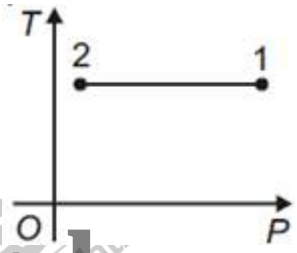
(3) 23 m/s

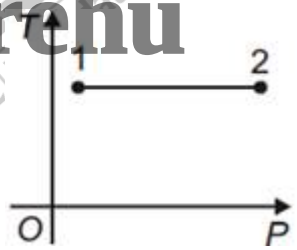
(4) 2.3 m/s

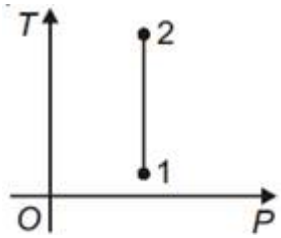
28. Consider pressure versus volume (*P*-*V*) diagram for an ideal gas. Corresponding temperature versus pressure (*T*-*P*) graph is



(1)


(2)


(3)


(4)


29. The average speed of the molecules of a homonuclear diatomic gas is 2 km/s. When the absolute temperature is doubled, the molecules dissociate into two atoms. The new average speed of the atom is

(1) 4 km/s

(2) 1 km/s

(3) $\sqrt{2}\text{km/s}$

(4) $2\sqrt{2}\text{km/s}$

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30. One mole of an ideal gas undergoes a process $P = \frac{P_0}{1+\frac{V_0}{V}}$, here P_0 and V_0 are constants. Change in temperature of the gas when volume is changed from $V = V_0$ to $V = 2V_0$ is

(1) $\frac{6P_0V_0}{5R}$

(2) $\frac{5P_0V_0}{6R}$

(3) $-\frac{6P_0V_0}{5R}$

(4) $-\frac{5P_0V_0}{6R}$

31. A vessel is filled with oxygen gas at temperature T K. Imagine an event described below. Two oxygen molecules each of mass "m", moving with equal speed "V" in opposite directions collide with each other. Subsequently, they stop momenteirely and the lost energy is used to dissociate the two molecules at the same temperature. The value of "V" is

(K_B = Boltzmann's constant)

(1) $V = \sqrt{\frac{8K_BT}{m}}$

(2) $V = \sqrt{\frac{2K_BT}{m}}$

(3) $V = \sqrt{\frac{K_BT}{m}}$

(4) $V = 2\sqrt{\frac{K_BT}{m}}$

32. Consider the following statements about kinetic theory of gases:
(A) The total kinetic energy of a gas is independent of its temperature.
(B) The rms speed of gas molecules increases with increase in temperature.
(C) All molecules of a gas move with the same velocity at a given temperature
Which of the above statement(s) is/are correct?

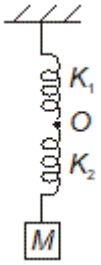
(1) Both (A) and (B)

(2) Only (B)

(3) Only (C)

(4) Both (B) and (C)

33. A block of mass *M* starts to oscillate in SHM. If amplitude of block is *Y* then amplitude of point *O* is



(1) $\frac{K_1Y}{K_2}$

(2) $\frac{K_1Y}{K_1+K_2}$

(3) $\frac{K_2Y}{K_1+K_2}$

(4) $\frac{K_2Y}{K_1}$

34. The phase difference between velocity and acceleration of simple harmonic oscillator is

(1) π

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{3}$

(4) $\frac{\pi}{2}$

35. A simple pendulum performs SHM about $x = 0$ with an amplitude *A* and time period *T*. The speed of the pendulum at $x = \frac{3A}{4}$ will be

(1) $\frac{\sqrt{3}\pi A}{T}$

(2) $\frac{A\pi\sqrt{7}}{2T}$

(3) $\frac{\pi A\sqrt{5}}{2T}$

(4) $\frac{\pi A\sqrt{7}}{4T}$

36. When a particle executes two mutually perpendicular simple harmonic motions simultaneously, the path of the particle is

(1) A straight line

(2) A circle

(3) An ellipse

(4) Any of these depending on the phase difference and amplitude

37. The time period of particle executing S.H.M. according to the equation $y = (\sin\omega t - \cos\omega t)$, is

(1) $\frac{\omega}{2\pi}$

(2) $\frac{\omega}{\pi}$

(3) $\frac{\pi}{\omega}$

(4) $\frac{2\pi}{\omega}$

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38. A mass *m* attached to a spring oscillates with a time period 4 s. If the mass is increased by 1 kg, then period increases by 1 s. Its initial mass is

(1) $\frac{16}{9}$ kg

(2) $\frac{4}{3}$ kg

(3) $\frac{8}{9}$ kg

(4) $\frac{4}{9}$ kg

39. Out of the following functions representing motion of a particle which represents SHM?

(A) $y = \sin\omega t - \cos\omega t$

(B) $y = \sin^3\omega t$

(C) $y = 5\cos\left(\frac{3\pi}{4} - 3\omega t\right)$

(D) $y = 1 + \omega t + \omega^2 t^2$

(1) Only (A) and (B)

(2) Only (A)

(3) Only (D) does not represent SHM

(4) Only (A) and (C)

40. A progressive wave is represented by $y = 5\sin(100\pi t - 2\pi x)$ where *x* and *y* are in metre and *t* is in second. The maximum particle velocity is

(1) 100π m/s

(2) 200π m/s

(3) 400π m/s

(4) 500π m/s

41. The second overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe *L* meter long. The length of the open pipe will be

(1) *L*

(2) 2*L*

(3) 4*L*

(4) $\frac{L}{2}$

42. The transverse displacement of a wave pulse on a string is given by $y = \frac{0.4}{(16x^2 + 25t^2 + 2 + 40xt)}$. If *x* and *y* are in meter and *t* in second then distance covered by pulse in 2 s is

(1) 0.75 m

(2) 2.5 m

(3) 1.25 m

(4) 0.25 m

43. The speed of sound is maximum in

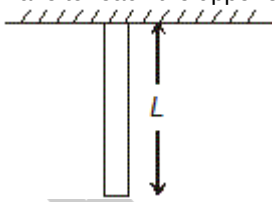
(1) Air

(2) Water

(3) Steel

(4) Vacuum

44. A thick uniform rope of length *L* is hanging from a rigid support as shown. A transverse wave is sent from the lower end of the rope towards upper end. The time taken by the wave to reach the upper end from lower end will be.



(1) $\sqrt{\frac{2L}{g}}$

(2) $\sqrt{\frac{L}{g}}$

(3) $\sqrt{\frac{L}{2g}}$

(4) $2\sqrt{\frac{L}{g}}$

45. The equation of stationary wave along a stretched string is given by $y = 5\sin\left(\frac{\pi x}{6}\right)\cos 20\pi t$, where *x* and *y* are in cm. The separation between two adjacent antinodes is

(1) 6 cm

(2) 4 cm

(3) 2.5 cm

(4) 2 cm

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CHEMISTRY

46. The oxidation numbers of H in CaH_2 is

(1) 0

(2) +1

(3) $+\frac{1}{2}$

(4) –1

47. In which of the following processes entropy decreases?

(1) Sublimation of naphthalene

(2) Melting of ice

(3) Condensation of water vapour

(4) Denaturation of protein

48. For the given reaction $A(l) \rightarrow 2\,B(g);\ \Delta U = -4.5\ kcal$, $\Delta S = 12\ cal\ K^{-1}$ at 27°C.
Hence ΔG for the reaction is

(1) –5.4 kcal

(2) –8.5 kcal

(3) –6.9 kcal

(4) –12.5 kcal

49. Choose the **incorrect** statement among the following.

(1) The first ionisation enthalpy of element with atomic number 19 is lower than that of the element with atomic number 20

(2) Negative electron gain enthalpy of element with atomic number 17 is more than that of the element with atomic number 34

(3) Oxides of elements with atomic number 13 and 50 are amphoteric and acidic respectively

(4) Atomic radii of element with atomic number 31 is less than that of element with atomic number 13

50. Match List-I with List-II.

List-I (Atomic number)	**List-II (IUPAC official name)**
a. 101	(i) Lawrencium
b. 103	(ii) Bohrium
c. 104	(iii) Mendelevium
d. 107	(iv) Rutherfordium

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(ii), b(iii), c(iv), d(i)

51. Which of the following are correctly matched?

Elements	**Atomic number**	**Group number**	**Period number**
(A)	42	6	5
(B)	58	3	6
(C)	48	10	5
(D)	56	2	6

Choose the correct option.

(1) (A), (B) and (D) only

(2) (A), (C) and (D) only

(3) (B) and (C) only

(4) (B), (C) and (D) only

52. Match List-I with List-II.

List-I	**List-II**
(a) NO	(I) Acidic
(b) CO_2	(II) Basic
(c) ZnO	(III) Neutral
(d) CaO	(IV) Amphoteric

Choose the correct answer from the options given below.

(1) (a)(II), (b)(IV), (c)(I), (d)(III)

(2) (a)(III), (b)(I), (c)(II), (d)(IV)

(3) (a)(III), (b)(I), (c)(IV), (d)(II)

(4) (a)(I), (b)(III), (c)(IV), (d)(II)

53. Match the column I with column II

Column I (Element in given compound)	**Column II (Oxidation state)**
a. Central carbon atom in C_3O_2	(i) +6
b. Central bromine atom in Br_3O_8	(ii) +5
c. Chromium atom in CrO_5	(iii) +4
d. Terminal sulphur atom in $S_4O_6^{2-}$	(iv) 0

Choose the option containing correct match.

(1) a(iii), b(i), c(iv), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iii), b(iv), c(i), d(ii)

54. Which of the following is a combination redox reaction?

(1) $2\,KClO_3 \xrightarrow{\Delta} 2\,KCl\left(s\right) + 3O_2\left(g\right)$

(2) $Zn\left(s\right) + 2\,HCl\left(aq\right) \rightarrow ZnCl_2\left(aq\right) + H_2\left(g\right)$

(3) $CH_4\left(g\right) + 2O_2\left(g\right) \xrightarrow{\Delta} CO_2\left(g\right) + 2H_2O\left(l\right)$

(4) $Br_2\left(l\right) + 2I^- \rightarrow 2\,Br^-\left(aq\right) + I_2\left(s\right)$

55. Which of the following reactions are disproportionation reactions?

(a) $P_4\,(s) + 3\,OH^-\,(aq) + 3H_2O\,(l) \rightarrow PH_3\,(g) + 3H_2\,PO_2^-\,(aq)$

(b) $2\,KMnO_4 \xrightarrow{\Delta} K_2\,MnO_4 + MnO_2 + O_2$

(c) $S_8\,(s) + 12\,OH^-\,(aq) \rightarrow 4S^{2-}\,(aq) + 2S_2O_3^{2-}\,(aq) + 6H_2O\,(l)$

(d) $2\,MnO_4^- + 3\,Mn^{2+} + 2H_2O \rightarrow 5\,MnO_2 + 4H^+$

(e) $3\,Cl_2(g) + 6\,OH^-(aq) \rightarrow 5\,Cl^-(aq) + ClO_3^-(aq) + 3H_2O(l)$

Choose the correct option.

(1) (a), (b) and (c) only

(2) (c), (d) and (e) only

(3) (a), (c) and (e) only

(4) (a), (b), (d) and (e) only

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56. If the enthalpy of fusion of a compound at 127°C is 36.2 kJ/mol then the entropy change for the fusion is (The melting point of the compound is 127°C)

(1) $75.2\ JK^{-1}mol^{-1}$

(2) $90.5\ JK^{-1}mol^{-1}$

(3) $285\ JK^{-1}mol^{-1}$

(4) $40\ JK^{-1}mol^{-1}$

57. Oxidation numbers of A, B and C are +2, –1 and +1 respectively. Possible formula of the compound is

(1) $A_2(B_3C)_2$

(2) $A_3(BC_2)_2$

(3) $A_2(BC_2)_2$

(4) $A_2(B_2C)_2$

58. 2 moles of a gas is expanded isothermally and reversibly from 10 lit. to 50 lit at 300 K temperature. Work done during the process is (Given $\log_{10}5 = 0.7$)

(1) – 8.04 kJ

(2) + 12.5 kJ

(3) – 4.01 kJ

(4) + 6.2 kJ

59. Read the statements marked as assertion (A) and reason (R).
Assertion (A): For isothermal reversible change work done by the system is equal to change in internal energy.
Reason (R): Change in internal energy depends on temperature of system.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

60. Choose the **incorrect** statement among the following.

(1) No work is done during free expansion of an ideal gas

(2) (U + PV) is a state function

(3) $\Delta_fH°$ for $Br_2(\ell)$, C(diamond) and $Cl_2(g)$ is zero

(4) When a system is in equilibrium, the entropy is maximum and the change in entropy, $\Delta S = 0$

61. Given below are two statements:
Statement I: For the process, $H_2O(l)$ (1 bar, 373 K) ⇌ $H_2O(g)$ (1 bar, 373 K) ; $\Delta G < 0$ and $\Delta S > 0$.
Statement II: Both molar heat capacity and internal energy are extensive properties.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both statement I and statement II are incorrect

(2) Both statement I and statement II are correct

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

62. Two moles of an ideal gas undergo reversible isothermal expansion from 2.303 L to 23.03 L at 27°C. The change in entropy is ($R = 8.314\ J\ mol^{-1}K^{-1}$)

(1) $38.3\ J\ K^{-1}$

(2) $16.6\ J\ K^{-1}$

(3) $4.6\ J\ K^{-1}$

(4) $-4.6\ J\ K^{-1}$

63. If the enthalpies of formation of $C_3H_8(g)$, $CO_2(g)$ and $H_2O(l)$ at standard conditions are –104 kJ mol^{-1}, –394 kJ mol^{-1} and –286 kJ mol^{-1} respectively then the standard enthalpy of combustion of 66 g of $C_3H_8(g)$ will be

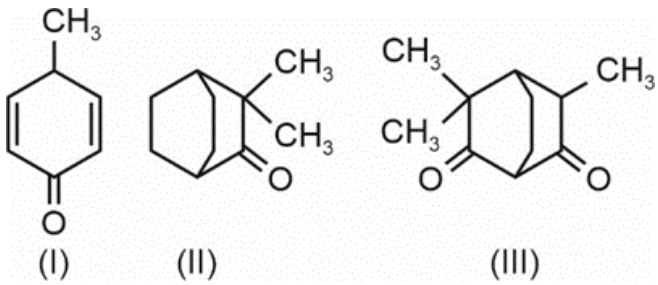
(1) –2222 kJ

(2) –2500 kJ

(3) –784 kJ

(4) –3333 kJ

64.



Which of the given compounds can exhibit tautomerism?

(1) I, II and III

(2) Only I and II

(3) Only I and III

(4) Only II and III

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65. In Carius method of estimation of halogen, 0.2 g of an organic compound gave 0.15 g of AgBr. The percentage of bromine in the compound will be (Atomic masses of Ag and Br respectively are 108 u and 80 u)

(1) 31.9%

(2) 62.5%

(3) 39.5%

(4) 26.2%

66. The incorrect statement is

(1) Glycerol can be separated from spent-lye in soap industry by distillation under reduced pressure.

(2) Chloroform and aniline can be separated by distillation.

(3) Fractions of crude oil in petroleum industry is separated by fractional distillation.

(4) Aniline cannot be separated from aniline water mixture by steam distillation.

67. Number of cyclic ethers excluding stereoisomers for molecular formula C_4H_8O is

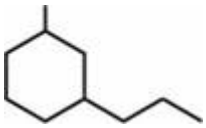
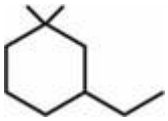
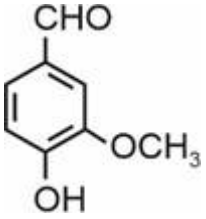
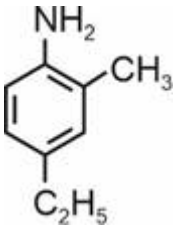
(1) 3

(2) 4

(3) 5

(4) 6

68. Choose the incorrect match.

	Organic compound	IUPAC name
(1)		1-Methyl-3-propylcyclohexane
(2)		1-Ethyl-3,3-dimethylcyclohexane
(3)		4-Hydroxy-3-methoxy benzaldehyde
(4)		4-Ethyl-2-methylaniline

(1) (1)

(2) (2)

(3) (3)

(4) (4)

69. Which of the following statement(s) is/are correct?
A. Hyperconjugation involves delocalisation of σ electrons of C – H bond of an alkyl group directly attached to an atom of unsaturated system.
B. Hyperconjugation involves delocalisation of σ electrons of C – H bond of an alkyl group directly attached to an atom with an unshared *p* orbital.
C. Hyperconjugation is a permanent effect.

(1) A only

(2) A and B only

(3) A and C only

(4) A, B and C

70. In Dumas method for estimation of nitrogen, 0.1 g of an organic compound gave 30 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is
(Aqueous tension at 300 K = 15 mm Hg)

(1) 19.7%

(2) 49.1%

(3) 28.2%

(4) 31.4%

71. Match column-I with column-II.

Column-I (Compounds)	Column-II (Type of isomerism)
a. Isopentane and neopentane	(i) Functional group isomerism
b. Propan-1-ol and propan-2-ol	(ii) Position isomerism
c. Propanone and propanal	(iii) Metamerism
d. Methoxypropane and ethoxyethane	(iv) Chain isomerism

Choose the correct answer from the options given below.

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(iv), b(iii), c(i), d(ii)

72. In the following reaction

$CH_3COONa + NaOH \xrightarrow[\Delta]{CaO}$ (A) Major + Na_2CO_3

The incorrect statement about (A) is

(1) On heating with O_2 in the presence of Mo_2O_3, gives methanoic acid

(2) In the presence of Cu, it reacts with dioxygen at 523 K and 100 atm gives methanol

(3) It reacts with steam at 1273 K in the presence of Ni catalyst to form carbon monoxide and dihydrogen

(4) On heating in the presence of air, it is completely oxidized to carbon dioxide and water

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73. If 200 ml of 0.1 M H_2SO_4 solution is mixed with 200 ml of 0.15 M NaOH solution then heat evolved during the neutralization process is

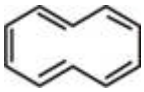
(1) 411 cal

(2) 548 cal

(3) 959 cal

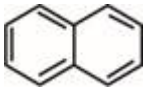
(4) 822 cal

74. Aromatic compound among the following is

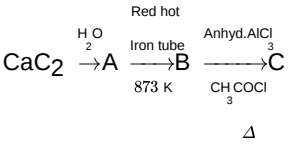
(1) 

(2) 

(3) 

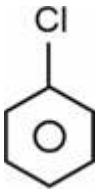
(4) 

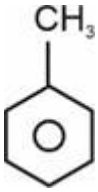
75. Consider the following sequence of reaction.

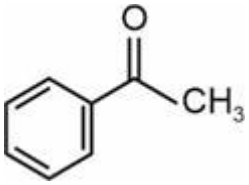


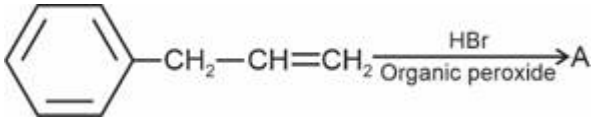
Major product C is

(1) 

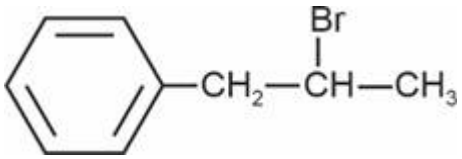
(2) 

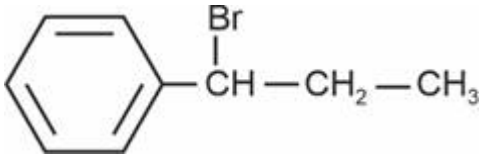
(3) 

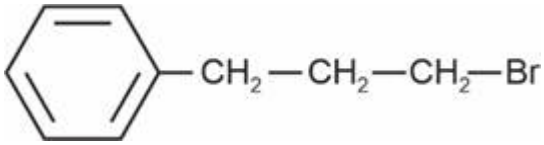
(4) 

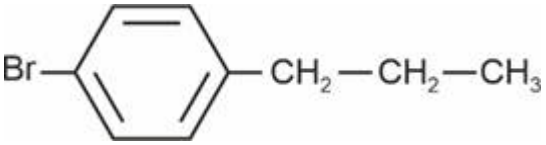
76. 

The major product A in above mentioned reaction is

(1) 

(2) 

(3) 

(4) 

77. In the following reaction



Incorrect statement about product (C) is

(1) It is less acidic than ethane

(2) It reacts with water on warming with mercuric sulphate and dilute sulphuric acid at 333 K to form ethanal

(3) It reacts with two molecules of hydrogen bromides to form gem dibromides

(4) It reacts with $NaNH_2$ to liberate NH_3 gas

78. Appearance of blood red colour takes place in Lassaigne test for the quantitative analysis of an organic compound that contains

(1) Both sulphur and bromine

(2) Both sulphur and nitrogen

(3) Nitrogen only

(4) Sulphur only

79. Which among the following processes is an endothermic process?

(1) $F(g) \rightarrow F^-(g)$

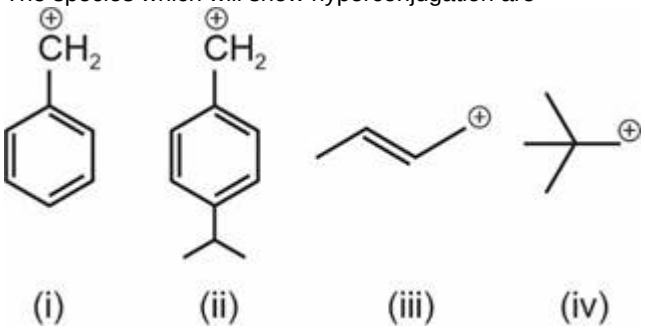
(2) $Br(g) \rightarrow Br^-(g)$

(3) $O^-(g) \rightarrow O^{2-}(g)$

(4) $O(g) \rightarrow O^-(g)$

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80. The species which will show hyperconjugation are



(1) (i), (ii) and (iii) only

(2) (ii), (iii) and (iv) only

(3) (ii) and (iii) only

(4) (i) and (iii) only

81. Match List-I with List-II.

List-I (Compound)	List-II (Number of σ-bonds)
(a) Anthracene	(i) 23
(b) Biphenyl	(ii) 18
(c) Gammaxane	(iii) 19
(d) Naphthalene	(iv) 26

Choose the correct match

(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(2) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

82. Consider the following statements
(a) Gallium is also known as Eka-silicon
(b) Fourth period of modern periodic table contains 18 elements
(c) The elements with atomic number more than 92 are called transuranium elements
The correct statements are

(1) (b) and (c) only

(2) (a) and (b) only

(3) (a) and (c) only

(4) (a), (b) and (c)

83. Reaction for which ΔH is equal to ΔU, is

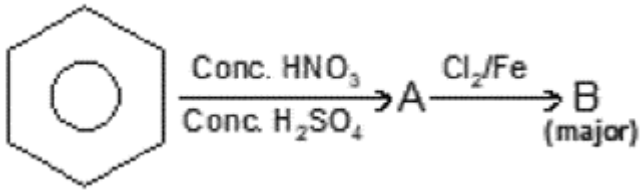
(1) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

(2) $C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$

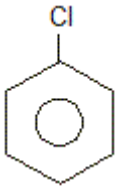
(3) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

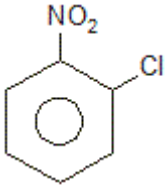
(4) $C(s) + O_2(g) \rightarrow CO_2(g)$

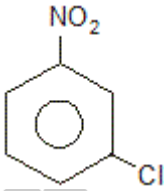
84.



Product B is

(1) 

(2) 

(3) 

(4) 

85. Total number of geometrical isomers possible in penta-1,3-diene is

(1) 4

(2) 6

(3) 2

(4) 8

86. Consider the following statements
(a) Cl exhibits its highest oxidation state in $NaClO_3$.
(b) Conversion of $Cr_2O_7^{2-}$ to CrO_4^{2-} does not require a reducing agent.
(c) Cl exists in both positive and negative oxidation state in $CaOCl_2$.
Correct statement(s) is/are

(1) (c) only

(2) (a), (b) and (c)

(3) (b) and (c) only

(4) (a) only

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87. Given below are the two statements:
Statement I: Dipole moment of cis-but-2-ene is higher than the dipole moment of trans-but-2-ene.
Statement II: Boiling point of trans-but-2-ene is higher than that of cis-but-2-ene.
In light of the above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

88. Correct order of boiling point of the given compounds is
(a) Butane
(b) 2, 2-Dimethylpropane
(c) 2-Methylbutane
(d) 2-Methylpropane

(1) (b) > (a) > (c) > (d)

(2) (c) > (b) > (a) > (d)

(3) (c) > (a) > (b) > (d)

(4) (b) > (c) > (d) > (a)

89. Consider the following reaction
$aKMnO_4 + bHCl \rightarrow cKCl + dMnCl_2 + eCl_2 + fH_2O$
Coefficients a, b and e for balanced chemical reaction respectively are

(1) 2, 16 ,5

(2) 1, 8, 4

(3) 2, 7, 5

(4) 1, 7, 3

90. Given below are two statements:
Statement I: Ethyl fluoride on reaction with zinc and dilute HCl gives ethane as major product.
Statement II: Ethyl bromide on reaction with sodium in presence of dry ether gives n-butane as major product.
In light of the above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

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91. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Bivalents clearly visible as tetrad	(i)	Diakinesis
b.	Chiasmata disappear	(ii)	Pachytene
c.	Nucleolus starts to disappear	(iii)	Zygotene
d.	Synapsis of chromosomes occurs	(iv)	Prophase

(1) a(iii), b(ii), c(i), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(i), b(ii), c(iii), d(iv)

92. Select the events which occur twice during meiosis.
a. Nuclear division
b. Division of cytoplasm
c. DNA replication
d. Bivalent formation
e. Equational division

(1) a and b only

(2) a, b and e

(3) a, b and d

(4) c, d and e

93. In oocytes of some vertebrates, which of the following stages in meiosis can last for years?

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) Diplotene

94. Quiescent or G_0 stage of cell is

(1) Metabolically inactive stage

(2) Not seen in plant cells

(3) The stage where cells do not proliferate unless they are called to do so

(4) Part of M phase

95. How many of the following events is/are associated with S phase?
(a) DNA replication
(b) Tubulin protein synthesis
(c) Duplication of mitochondria
(d) Duplication of centrioles

(1) One

(2) Two

(3) Three

(4) Four

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96. The phase between two successive M phases is called

(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Interphase

97. Observe the following events:
a. Formation of the site of active ribosomal RNA synthesis.
b. Formation of the cell plate in the centre to form two daughter cells.
c. Complete disappearance of perinuclear space.
d. Centrosome moving towards the pole.
e. Attachment of spindle fibres to the sites found on the primary constriction.
f. Splitting of primary constriction.
Select the option depicting the **correct** sequence of events that occur during the formation of liquid endosperm in coconut.

(1) d → c → e → f → a → b

(2) c → e → f → d → b

(3) e → f → d → c → a

(4) c → e → f → a

98. End of cell division is marked by

(1) Karyokinesis

(2) Telophase

(3) Cytokinesis

(4) Interphase

99. Read the following statements and choose the **correct** option.
a. *nigrum*, *melongena* and *tuberosum* are the specific epithets of genus *Solanum*.
b. *Felis* and *Panthera* have more common characters than *Felis* and *Canis*.
c. Monkey, gorilla and gibbon are placed in same order Mammalia.
d. Anacardiaceae is a family of dicot plants.

(1) Only a & b are incorrect

(2) Only a, b & c are correct

(3) a, b, c & d all are correct

(4) Only 'c' is incorrect

100. Find out the **incorrect** match

	Common name	Order	Class
A.	Housefly	Diptera	Insecta
B.	Mango	Sapindales	Dicotyledonae
C.	Wheat	Poales	Monocotyledonae
D.	Man	Mammalia	Primata

(1) A

(2) B

(3) C

(4) D

101. The process of categorising different organisms on the basis of some easily observable characters is called

(1) Nomenclature

(2) Identification

(3) Classification

(4) Taxonomy

102. Identify the **correctly** printed scientific name

(1) *Mangifera indica Linnaeus*

(2) *Mangifera Indica* Linnaeus

(3) *Mangifera indica* Linn.

(4) Mangifera Indica Linn

103. Given below are two statements **Assertion (A)** and **Reason (R)**. Choose the most appropriate answer for them.
Assertion (A): Internode of the shoot gets condensed and produce different floral appendages laterally at successive nodes instead of leaves.
Reason (R): Flower is a modified shoot where shoot apical meristem changes to floral meristem.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) (A) is true but (R) is false.

(3) Both (A) and (R) are false.

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A).

104. Select the **incorrect** match.

(1) Marginal placentation	–	Pea
(2) Axile placentation	–	Lemon
(3) Parietal placentation	–	*Dianthus*
(4) Basal placentation	–	Marigold

(1) (1)

(2) (2)

(3) (3)

(4) (4)

105. A compound leaf differs from a simple leaf in

(1) Bearing axillary buds in axil of its leaflets

(2) Having reticulate venation

(3) Having a number of leaflets as incisions of the lamina reach upto the midrib

(4) Bearing a bud in the axil of the petiole

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106. Examine the following structures & identify the haploid, diploid and triploid structures, respectively.
(a) Endosperm of castor
(b) Seed coat of pea
(c) Embryo sac of *Hibiscus*
(d) Aleurone layer of wheat
(e) Scutellum of maize
(f) Radicle of *Petunia*

	Haploid	Diploid	Triploid
(1)	(b), (e)	(c), (f)	(a), (d)
(2)	(c), (e)	(d)	(a), (b), (f)
(3)	(c)	(b), (e), (f)	(a), (d)
(4)	(a), (b)	(d), (e)	(c), (f)

(1) (1)

(2) (2)

(3) (3)

(4) (4)

107. Read the following statements and select the **correct** option.
Statement-A: The hilum is a scar on the seed coat.
Statement-B: A seed has protective covering called seed coat which is further differentiated into inner testa and outer tegmen layers.

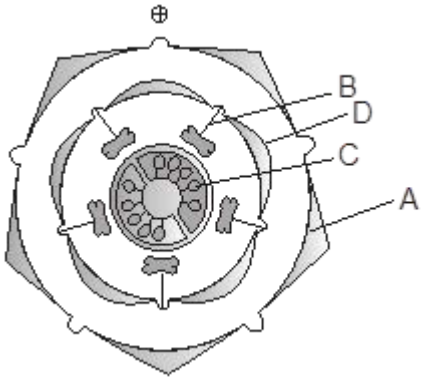
(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are correct

(4) Both statements are incorrect

108. Figure shows the floral diagram of 'makoi' with labels A to D. Select the option which gives **correct** identification of label with function or characteristic.



(1) C – Ovule – Free central placentation

(2) A – Sepals – Persistent

(3) B – Stamens – Polyadelphous

(4) D – Petals – Twisted aestivation

109. Identify the similar features between potato and mustard family that can be depicted in their floral formula.
(a) Bicarpellary, syncarpous gynoecium
(b) Epipetalous stamen
(c) Bisexual and actinomorphic flower
(d) Petals in a whorl just touch one another without overlapping
(e) More than three free sepals
How many of the above features are **true**?

(1) Two

(2) Three

(3) Four

(4) Five

110. In coconut, the mesocarp is

(1) Fibrous

(2) Edible

(3) Stony hard

(4) Outermost covering

111. Papilionaceous corolla is seen in

(1) Gram

(2) Tomato

(3) Chilli

(4) Lily

112. When calyx and corolla are not distinct then it is termed as perianth as in

(1) Tulip

(2) Mustard

(3) Bean

(4) *Petunia*

113. Epidermal tissue system comprises
a. Epidermal cells
b. Stomata
c. Trichomes
d. Hairs
e. Guard cells
The **correct** one(s) is/are

(1) a only

(2) a, b and e only

(3) All a, b, c, d and e

(4) a, b, c and d only

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114. State **true (T)** or **false (F)** in the following statements and select the **correct** option.
A. When bulliform cells are flaccid, they make the leaves curl inwards to minimize water loss.
B. In dicot leaves, the size of the vascular bundles is dependent on the size of the veins.
C. The adaxial epidermis bears more stomata than the abaxial epidermis in dicot leaves.

	A	B	C
(1)	T	F	T
(2)	F	F	T
(3)	T	T	F
(4)	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

115. Choose the **incorrect d**ifference between dicot stem and monocot root?

	Dicot stem	Monocot root
(1)	Endarch xylem	Exarch xylem
(2)	Conjoint and open vascular bundles	Radial and closed vascular bundles
(3)	Phloem parenchyma is present	Phloem parenchyma is absent
(4)	Pith is inconspicuous or absent	Pith is small but developed

(1) (1)

(2) (2)

(3) (3)

(4) (4)

116. Read the following statements (A - D)
(A) The outer wall of guard cells are thick while the inner walls (towards the stomatal pore) are very thin.
(B) Stomata regulates the process of transpiration and gaseous exchange.
(C) All the tissues on the inner side of the hypodermis constitute the stele.
(D) In Sunflower root, hypodermis is protective in function and pith is well-developed
In the light of above statements, select the option for **incorrect** one(s).

(1) Only (A) and (B)

(2) Only (A), (C) and (D)

(3) Only (C) and (D)

(4) Only (B)

117. Pericycle forms semi-lunar patches of sclerenchyma cells in

(1) Dicot roots

(2) Monocot stems

(3) Dicot stems

(4) Monocot roots

118. The specialised epidermal cells present in the vicinity of guard cells are called

(1) Bulliform cells

(2) Companion cells

(3) Subsidiary cells

(4) Endodermal cells

119. Read the given assertion (A) and reason (R) and choose the **correct** option.
Assertion (A) : In stems of grasses, large sized vascular bundles are arranged in a ring-like manner.
Reason (R) : The large sized vascular bundles in monocot stem are densely arranged towards the periphery as compared to smaller ones.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

120. In dicot stem, hypodermis is made up of

(1) Parenchyma

(2) Collenchyma

(3) Sclerenchyma

(4) Sclereids

121. Match the columns and select the **correct** option.

	Column I		Column II
a.	Auxin	(i)	Stem elongation in cabbage
b.	Ethylene	(ii)	Delay the ageing of leaves
c.	Gibberellin	(iii)	Prevents abscission of young flowers and fruits
d.	Cytokinin	(iv)	Breaks dormancy of buds & seeds

(1) a(i), b(iii), c(iv), d(ii)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(ii), c(i), d(iv)

122. Abscisic acid is also known as stress hormone because

(1) It opens the stomata present on stem

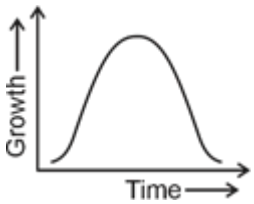
(2) Its synthesis is stimulated by adverse environmental conditions

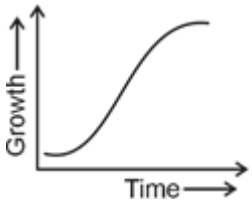
(3) It promotes loss of water through leaves and increases water stress

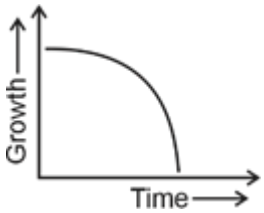
(4) It inhibits dormancy of seeds

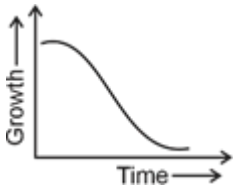
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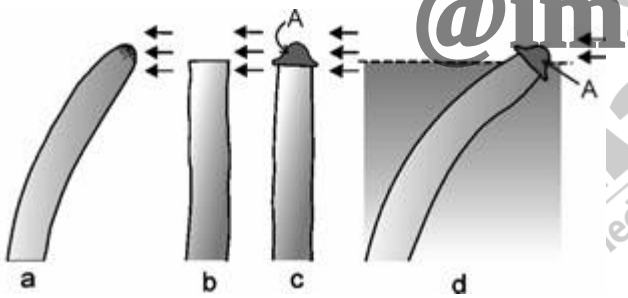
123. The growth curve which is characteristic of living organisms growing in a natural environment can be represented as

(1) 

(2) 

(3) 

(4) 

124. 

Which of the following conclusions **cannot** be drawn from the above given diagram of oat coleoptile upon unilateral illumination?

(1) If label A represents an opaque cover on coleoptile tip, then it inhibits bending of coleoptile.

(2) Presence of coleoptile tip is not essential for bending.

(3) If label A represents transparent cover on the coleoptile tip, then the bending of coleoptile will occur.

(4) Unilateral illumination of a decapitated coleoptile will not cause bending.

125. Zygotic divisions initially

(1) Show arithmetic growth

(2) Can be expressed by a straight line

(3) Show a pattern where every cell divides including all the daughter cells

(4) Has only one cell dividing whereas other cells undergo differentiation

126. Adenine derived plant growth regulator can

(1) Induce apical dominance

(2) Be used to kill dicotyledonous weed

(3) Promote bolting in plant

(4) Delay leaf senescence

127. Application of which of the given plant hormones in cucumber increases the number of female flowers?

(1) IAA

(2) GA_3

(3) Kinetin

(4) Ethylene

128. Rapid internode/petiole elongation in deep water rice plants is promoted by

(1) Ethylene

(2) Abscisic acid

(3) Cytokinin

(4) Auxin

129. When apical bud of a plant is removed then

(1) Auxin shows apical dominance

(2) Cytokinin is removed along with auxin

(3) More lateral branches will be seen

(4) Root formation is promoted

130. Efficiency index is the measure of

(1) Ability of the plant to produce new plant material

(2) Time of growth

(3) Number of cells that lose capacity to divide

(4) Rate of differentiation of plant cells

131. In the geometrical growth of microorganisms, the initial phase is

(1) Exponential phase

(2) Lag phase

(3) Stationary phase

(4) Log phase

132. Heterophyllous development in buttercup is due to

(1) Variation in temperature

(2) Availability and non-availability of nutrients

(3) Difference in surrounding environment

(4) Presence and absence of light

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133. Development is the sum of two processes that are

(1) Growth and differentiation

(2) Differentiation and dedifferentiation

(3) Differentiation and redifferentiation

(4) Cell division and senescence

134. Select the **odd** one w.r.t. synthetic auxins

(1) 2,4-D

(2) IBA

(3) NAA

(4) 2,4,5-T

135. Miller et. al identified and crystallised the cytokinesis promoting active substance named

(1) Kinetin

(2) ABA

(3) IAA

(4) Adenine

ZOOLOGY

136. Read the four statements (a-d) given below.
a. Bundle of His is present in the pericardium.
b. P–wave in ECG represents atrial depolarisation.
c. Single circulation is found in fishes and amphibians.
d. Maximum blood enters from atria into ventricles during ventricular diastole.
The two **correct** statements are

(1) a and b

(2) b and c

(3) a and d

(4) b and d

137. Read the following reactions and identify A and B respectively.

Prothrombin $\xrightarrow{A,\ Ca^{2+}}$ Thrombin

Fibrinogen $\xrightarrow{B,\ Ca^{2+}}$ Fibrin

Choose the **correct** option.

(1) Thrombokinase, Thrombin

(2) Thromboplastin, Thrombokinase

(3) Thrombin, Thrombokinase

(4) Thrombokinase, Thromboplastin

138. Select the **incorrect** statement w.r.t. the nodal tissue of the heart.

(1) The entire nodal tissue, from sino-atrial node to the Purkinje fibres has the ability to generate action potentials.

(2) All parts of the nodal tissue generate the same number of action potentials per minute.

(3) Both sino-atrial and atrio-ventricular nodes are present in the right atrium.

(4) Sino-atrial node normally acts as the pacemaker.

139. Which of the following represents the **correct** pathway of blood flow in the human body?

(1) Vena cava → Left ventricle → Pulmonary vein

(2) Vena cava → Left ventricle → Pulmonary artery

(3) Pulmonary artery → Right ventricle → Pulmonary veins

(4) Pulmonary veins → Left ventricle → Aorta

140. Read the following statements:
(a) A special neural centre called cardio-vascular centre is present in the medulla oblongata.
(b) Sympathetic nerve endings which release adrenaline which accelerates the heart beat.
(c) Parasympathetic nerve endings release acetylcholine which decreases the speed of conduction of action potentials in the heart.
(d) Adrenal medullary hormones released during resting conditions cause a reduction in the strength of heart contraction and cardiac output.
How many of the above statements are **correct**?

(1) Two

(2) One

(3) Four

(4) Three

141. A : In humans, leukocytes are only nucleated cells present in the body.
R : Monocytes are the largest and most abundant type of leukocytes.
In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

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142. The first heart sound occurs due to
(a) Closure of the tricuspid valve
(b) Opening of the semilunar valve
(c) Closure of the bicuspid valve
(d) Closure of the semilunar valves.
Choose the **correct** option.

(1) (a) and (c)

(2) (b) and (c)

(3) (b) only

(4) (a) and (d)

143. In animals, the circulatory pattern 'P' is considered to be more advantageous than the circulatory pattern 'Q'. The flow of fluid can be precisely regulated in former. Pattern 'P' is usually found in

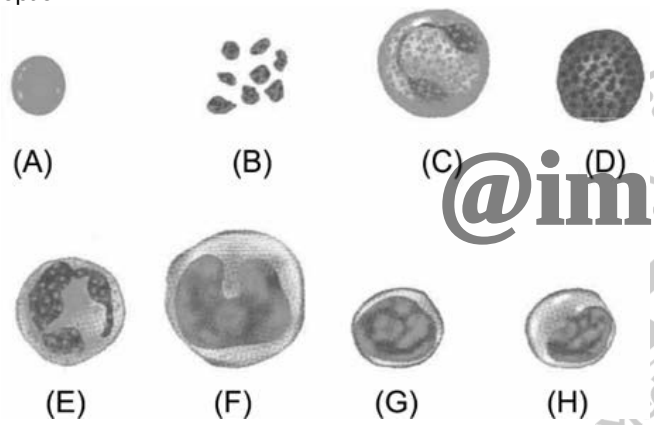
(1) Arthropods

(2) Fishes

(3) Most of molluscs

(4) Echinoderms

144. Observe the diagrammatic representation of formed elements in blood given below and choose the **correct** option



(1) (C) and (D) are nucleated agranulocytes

(2) (A) and (B) are non-nucleated red coloured cells

(3) (E) and (F) are nucleated phagocytic cells

(4) (G) and (H) are uninucleated granulocytes

145. If the cardiac output of a person is 4900 mL and blood volume in the ventricles at the end of diastole is 140 mL while at the end of ventricular systole is 70 mL. What is the heart rate of this person?

(1) 75 beats per minute

(2) 70 beats per minute

(3) 35 beats per minute

(4) 140 beats per minute

146. Choose the **mismatched** pair between the organism and its main excretory structure.

(1) Green glands - Prawns

(2) Malpighian tubules - Cockroach

(3) Nephridia - Earthworms

(4) Antennal glands - Leech

147. All of the following substances are eliminated by the sebaceous glands **except**

(1) Sterols

(2) Urea

(3) Hydrocarbons

(4) Waxes

148. Where are the specialised cells called podocytes found in a nephron?

(1) PCT

(2) DCT

(3) Bowman's capsule

(4) Collecting duct

149. The given flow chart represents feedback mechanism that regulates kidney function.

A in glomerular blood pressure/GFR

↓

Activates JG cells to release B

↓

B converts angiotensinogen to Angiotensin I

↓

Angiotensin II activates adrenal gland to release C

↓

D in glomerular blood pressure

Select the option that represents A, B, C and D **correctly**.

	A	B	C	D
(1)	Fall	Rennin	ANF	Rise
(2)	Rise	Aldosterone	ADH	Fall
(3)	Fall	Renin	Aldosterone	Rise
(4)	Rise	ANF	Renin	Fall

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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150. Select the **incorrect** statement from options given below.
(1) An adult human excretes, on an average, 1 - 1.5 ml of urine per day
(2) ANF is responsible for inhibiting the release of renin from JGA
(3) On an average, 25-30 gm of urea is excreted out per day
(4) Presence of glucose and ketone bodies in urine are indicative of diabetes mellitus

(1) (1)

(2) (2)

(3) (3)

(4) (4)

151. If PCT part of nephrons is removed from mammalian kidneys, then what will be the primary effect?

(1) Urine will become more diluted

(2) Urine will become more concentrated

(3) No change will occur in concentration of urine

(4) Urine formation will cease

152. Which of the following substances is completely reabsorbed in the nephron of a healthy adult person?

(1) Ammonia

(2) Urea

(3) Uric acid

(4) Glucose

153. The DCTs of many nephrons open into a straight tube called A. These tubes open into renal pelvis through B in C.
Choose the correct option.

	A	B	C
(1)	Collecting duct	Calyces	Medullary pyramids
(2)	Loop of Henle	Calyces	Pelvis
(3)	Loop of Henle	Medullary pyramids	Pelvis
(4)	Collecting duct	Medullary pyramids	Calyces

(1) (1)

(2) (2)

(3) (3)

(4) (4)

154. Select the **odd one** w.r.t. the axial skeleton

(1) Ear ossicles

(2) Cranium

(3) Sternum

(4) Clavicle

155. The cranial bone that has a depression called sella tursica which harbours pituitary gland is

(1) Temporal

(2) Sphenoid

(3) Ethmoid

(4) Occipital

156. During a skeletal muscle contraction, which ion is directly responsible for unmasking the myosin binding site on actin to enable cross bridge formation?

(1) K^+

(2) Na^+

(3) Mg^{2+}

(4) Ca^{2+}

157. Match Column-I and Column-II.

	Column-I		Column-II
(a)	Pivot joint	(i)	Elbow
(b)	Hinge joint	(ii)	Between atlas and axis
(c)	Ball and socket	(iii)	Humerus and pectoral girdle
		(iv)	Knee

Choose the **correct** option.

(1) (a)-(iv); (b)-(ii); (c)- (iii)

(2) (a)-(ii); (b)-(i); (c)- (iii)

(3) (a)-(i); (b)-(iv); (c)- (ii)

(4) (a)-(iii); (b)-(iv); (c)- (i)

158. Select the **correct** match w.r.t. having the same number of bones in the human body.

(1) Cranial bones – Facial bones

(2) Carpals of one limb – Tarsals of one limb

(3) Cranial bones – Carpals of one wrist

(4) Vertebrae – Skull bones

159. Read the given features.
a. They are smooth in appearance
b. Presence of nucleus
c. Syncytial appearance
d. Presence of gap junctions
Choose the option that reflects features of muscle fibres located in the walls of the intestine.

(1) a and b only

(2) c and d only

(3) a, b and d

(4) a, b, c and d

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160. Select an incorrect statement w.r.t. ribs of human beings.

(1) Vertebral ribs are connected ventrally to the sternum.

(2) Each rib is a flat bone.

(3) Each rib is connected dorsally to the vertebral column.

(4) It has two articulation surfaces on its dorsal end.

161. A is a large, triangular, flat bone situated in the dorsal part of the thorax between the second and the seventh ribs in humans.
Select the option that fills the above blank **correctly**.

(1) Coxal bone

(2) Clavicle

(3) Scapula

(4) Sternum

162. Functions of the skeletal system include all of the following, **except**

(1) Erythropoiesis

(2) Locomotion

(3) Storage of minerals

(4) Production of body heat

163. In a resting muscle, the central part of thick filament in a sarcomere is not overlapped by thin filament. This central part is called

(1) H – zone

(2) Z – line

(3) I– band

(4) A – band

164. The part of the forebrain in humans which controls the urge for eating, satiety and drinking is the

(1) Amygdala

(2) Hypothalamus

(3) Epithalamus

(4) Thalamus

165. Multipolar neurons have

(1) One axon and two or more dendrites

(2) One dendrite only

(3) One axon and one dendrite

(4) Multiple dendrites only

166. Ions essential for depolarisation of axolemma upon receiving stimulus are

(1) Sodium

(2) Potassium

(3) Calcium

(4) Magnesium

167. Electrical synapses

(1) Involve neurotransmitter molecules like dopamine

(2) Permit only unidirectional movement of biomolecules

(3) Allow nerve impulses to travel faster than in a chemical synapse

(4) Work through tight junctions that allow the passage of ions

168. Which of the following is **not** a part of brain stem?

(1) Pons varolii

(2) Cerebellum

(3) Midbrain

(4) Medulla oblongata

169. During the resting state of a nerve fibre, the axonal membrane is

(1) More permeable to K^+ than Na^+

(2) More permeable to Na^+ than K^+

(3) Equally permeable to Na^+ and K^+

(4) More permeable to Cl^- and Na^+

170. Which of the following is present at the node of Ranvier in a neuron of the CNS?

(1) Myelin sheath

(2) Axolemma

(3) Schwann cell

(4) Oligodendrocyte

171. The effect of GnRH produced by the hypothalamus directly stimulates the

(1) Synthesis and secretion of gonadotrophins

(2) Secretion of milk in mammary glands

(3) Synthesis of carbohydrates

(4) Fetal ejection reflex

172. Under the influence of which hormone, interstitial cells in testes secrete androgens?

(1) FSH

(2) LH

(3) GH

(4) ACTH

173. Symptoms such as stunted growth, mental retardation, a low metabolic rate and deaf-mutism in a child indicate that the pregnant mother suffered from hyposecretion of

(1) Parathyroid gland

(2) Thyroid gland

(3) Pancreas

(4) Liver

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174. Complete the analogy w.r.t. source gland of the hormone.
GnRH : Hypothalamus : : MSH :_____

(1) Zona fasciculata

(2) Pars intermedia

(3) Zona reticularis

(4) Neurohypophysis

175. Select the **incorrect** statement about atrial natriuretic factor (ANF).

(1) It is a peptide hormone acting as a vasodilator.

(2) It is synthesized from atrial wall of the heart.

(3) It increases blood pressure to increase the GFR.

(4) It works antagonistically to renin-angiotensin-aldosterone system.

176. Choose the **incorrect** option w.r.t. the gland, its hormone and its principal action

	Gland	Hormone	Principal Action
(1)	Adrenal cortex	Adrenaline	Increases the breakdown of lipids, increases O_2 consumption, dilates pupil
(2)	Pancreas	Insulin	Stimulates transport of glucose from blood to muscles and adipose cells
(3)	Thymus	Thymosin	Stimulating effect on entire immune system and promotes maturation of T-lymphocytes
(4)	Thyroid	Calcitonin	It is involved in calcium homeostasis, acts as hypocalcemic hormone

(1) (1)

(2) (2)

(3) (3)

(4) (4)

177. A lobular structure located between the lungs, behind the sternum on ventral side of the aorta in the human body is

(1) Pineal gland

(2) Pancreas

(3) Thymus

(4) Adrenal gland

178. Read the following statements and choose the **correct** option.
Statement-A : Corticosteroids like cortisol are used in patients undergoing organ transplant to avoid graft rejection.
Statement-B : Cortisol produces anti inflammatory reactions and suppresses the immune response.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

179. Match the following column A and column B and choose the option with all **correct** matches

	Column-A		Column-B
a.	ADH	(i)	Pituitary
b.	ACTH	(ii)	Formation of RBC's
c.	Erythropoietin	(iii)	Diabetes mellitus
d.	Insulin	(iv)	Diabetes insipidus
e.	Adrenaline	(v)	Piloerection

(1) a(iv), b(i), c(ii), d(iii), e(v)

(2) a(i), b(iv), c(iii), d(ii), e(v)

(3) a(v), b(iv), c(iii), d(ii), e(i)

(4) a(i), b(ii), c(iii), d(iv), e(v)

180. Several groups of neurosecretory cells which produce hormones to regulate the synthesis and secretion of hormones of adenohypophysis, are called

(1) Nerve plexuses

(2) Neuroglia

(3) Nuclei

(4) Ganglia

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